

Sean Meehan
Secret-Level Security Clearance | April 2024-Present
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Objective

A master's student in applied mathematics at Tulane University seeking a position where I can combine and apply my experience in advanced mathematics and computer science.

Education

Master of Science in Applied Mathematics

Tulane University

Aug 2025- Present
New Orleans, LA

GPA: 4.0

- Relevant Courses: Applied Mathematics I, Scientific Computing II, Real Analysis I, Data Science, Probability II, Applied Mathematics II, Linear Models
- Research: Surveyed literature concerning iterative methods for solving sparse linear systems culminating in a comparative analysis of the Jacobi, Gauss-Seidel, Weighted Jacobi, Scheduled Over Relaxation (SOR), Scheduled Relaxation Jacobi, and Adaptive SOR algorithms.

Bachelor of Science in Mathematics, Minor in Computer Science

Samford University

Aug 2024- May 2025
Birmingham, AL

GPA: 3.94

- Honors: Full scholarship, Dean's list (every semester), Summa Cum Laude
- Relevant Courses: AI and Complex Computing Methods, Programming in Java
- Research: Used modern techniques to analyze graph representations of the human brain to study efficient network architectures and how they develop in the biological context with the hope of applying this knowledge to machine learning networks.

Birmingham-Southern College

Aug 2021- May 2024
Birmingham, AL

GPA: 3.91

- Honors: Full scholarship, Louise Hall and Robert W. Echols Mathematics Award and Stipend (received twice), CRC Chemistry Award, Dean's list (every semester)
- Relevant Courses: Programming Languages, Data Structures and Algorithms, Intro to Programming in Python, Linear Algebra, Differential Equations, Statistics, Calculus I-III, Complex Analysis, Abstract Algebra
- Research: Conducted in-depth research in Linear Algebra, Abstract Algebra, and Complex Analysis, resulting in research papers on Singular Value Decomposition, RSA Cryptography, and Fourier Analysis. Additionally, developed a high-level Python module for advanced Linear Algebra applications.

Skills

- Programming: Python, MATLAB, Java, Bash, C, Go
- Software: Docker, Git, VMWare Fusion, Unreal Engine, Office 365, Wireshark
- Mathematics: Differential Equations, Linear Algebra, Calculus, Statistics, Complex Analysis, Abstract Algebra
- Data Science: AI/ML, Google Colab, Jupyter Notebooks, Pandas, Matplotlib
- Research: Technical Reading and Writing
- Problem Solving: Mathematical and Programmatic Reasoning
- Communication: Teaching, Tutoring, Public Speaking, Leadership

Employment History

Cybersecurity and AI Research Assistant

May 2023-Oct 2025
Huntsville, AL

UAH Center for Cybersecurity Research and Education

- Engineered and deployed machine learning algorithms for cyber-defense systems using Keras, PyTorch, TensorFlow, and scikit-learn.
- Developed Python and Bash scripts to automate cyber-attack simulations, generating high-quality datasets for intrusion detection models.
- Orchestrated virtualized environments with VMWare Fusion and Linux, integrating Wireshark and other network analysis tools for data collection and preprocessing.
- Conducted in-depth statistical analysis and data visualization using NumPy, pandas, and Matplotlib to extract meaningful insights from cybersecurity datasets.
- Worked in an Agile environment to co-develop software development kits to model and simulate real-world false data injection attacks, enhancing security research initiatives.
- Integrated Docker containers with interactive visualization software to create dynamic, real-time attack scenario representations.
- Synthesized and compiled research findings in machine learning, cybersecurity, and advanced mathematics to contribute to the academic and professional community.

Graduate Teaching Assistant

Aug 2025-Present
New Orleans, LA

Tulane University

- Strengthened oral communication and instructional skills via regular lectures and student interaction.
- Gained proficiency in presenting complex material to individuals with varying degrees of understanding.
- Built strong professional communication skills through regular collaboration with students and faculty.

Private Tutor

Jan 2023-Present
Birmingham, AL

Self-employed

- Established and maintained a steady network of clientele through effective communication, tailored instruction, and consistent student success.
- Provided personalized tutoring in chemistry, physics, and advanced mathematics, guiding high school and college students toward academic excellence.
- Strengthened technical communication and leadership skills by adapting teaching strategies to diverse learning styles.
- Developed extensive mastery of the fundamentals of mathematics and physics via continuous teaching and problem-solving exercises.

Extracurriculars

Birmingham-Southern College Men's Soccer

- Player 2021-2023
- SAA Conference Champion 2022, NCAA National Tournament Participant 2022

Birmingham-Southern College Chess Club

- Founder and President